



IIT ROORKEE



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H-O-S MODEL IN TRADE AND INCOME DISTRIBUTION

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From Classical to Heckscher-Ohlin Trade Theory

- **Problems with the Ricardian Model**
 - *Strong Specialization/Discontinuous Adjustment*
 - *Indeterminacy of Final Terms of Trade*
 - *No Income Distribution Effects*
 - *Problems with the Labor Theory of Value*
 - *Demand* is an important determinant of value
 - *Other factors of production* are important (at least proximately) in the production of value.



From Classical to Heckscher-Ohlin Trade Theory

- **Increasing Opportunity Cost**
 - A “bowed out” (*concave*) production frontier
 - This will yield a continuous price-output relationship.
- **Neoclassical Value Theory**
 - With increasing opportunity cost, we will need demand to determine autarky equilibrium price
 - Demand also resolves the ToT indeterminacy



The Hecksher-Ohlin-Samuelson (HOS) Model

- Paul Samuelson, who pioneered the formalization of trade theory, developed a simple formal analysis of the HO theory, which is commonly called the HOS model:
 - 2 final goods: *Bread* and *Steel*;
 - 2 factors of production: *Capital* and *Labor*; and
 - 2 countries: *US* and *Canada*.



H-O-S In General

- Heckscher, and his student Ohlin, worked in the early part of the 20th century
- Samuelson refined their work after WWII
- Closer attention is paid in this model to each country's resource endowment



H-O-S Assumptions

- 2 countries
- 2 commodities
- 2 factors (labor and capital)
- Perfect competition exists in all markets
- Each country's endowment of factors is fixed
- Factors are mobile internally, but immobile internationally



H-O-S Assumptions (cont'd)

- Each producer has a wide range of options as to how to produce X or Y
 - if K is cheap relative to labor, a relatively capital-intensive method will be adopted
 - if K is expensive relative to labor, a relatively labor-intensive method will be adopted
- Each country has the same CRTS technology



Concepts and Terminology

- The capital-labor ratio for good X is simply K^X/L^X , and for Y is K^Y/L^Y
- If $K^X/L^X > K^Y/L^Y$, production of good X is capital intensive relative to production of good Y
- Also, production of Y must be relatively labor intensive (If $K^X/L^X > K^Y/L^Y$, then $L^Y/K^Y > L^X/K^X$)

Concepts and Terminology

- Country A is said to be capital abundant relative to Country B if $(K/L)^A > (K/L)^B$
- In such a case, Country B must be relatively labor abundant
- So:
 - **goods** are produced relatively K or L intensively
 - **countries** are relatively K or L abundant



Concepts and Terminology

- The factor price of labor (the wage) is denoted w
- The factor price of capital is denoted r
- If labor is relatively expensive, w/r will be a relatively big number
- If labor is relatively cheap, w/r will be a relatively small number



More on Factor Prices

- What makes labor relatively expensive?
 - If it is relatively scarce
- What makes labor relatively cheap
 - If it is relatively abundant
- So: If $(K/L)^A$ is a relatively big number (that is, capital is relatively abundant), w/r will be a relatively big number, reflecting the relative scarcity of L and abundance of K



The Result

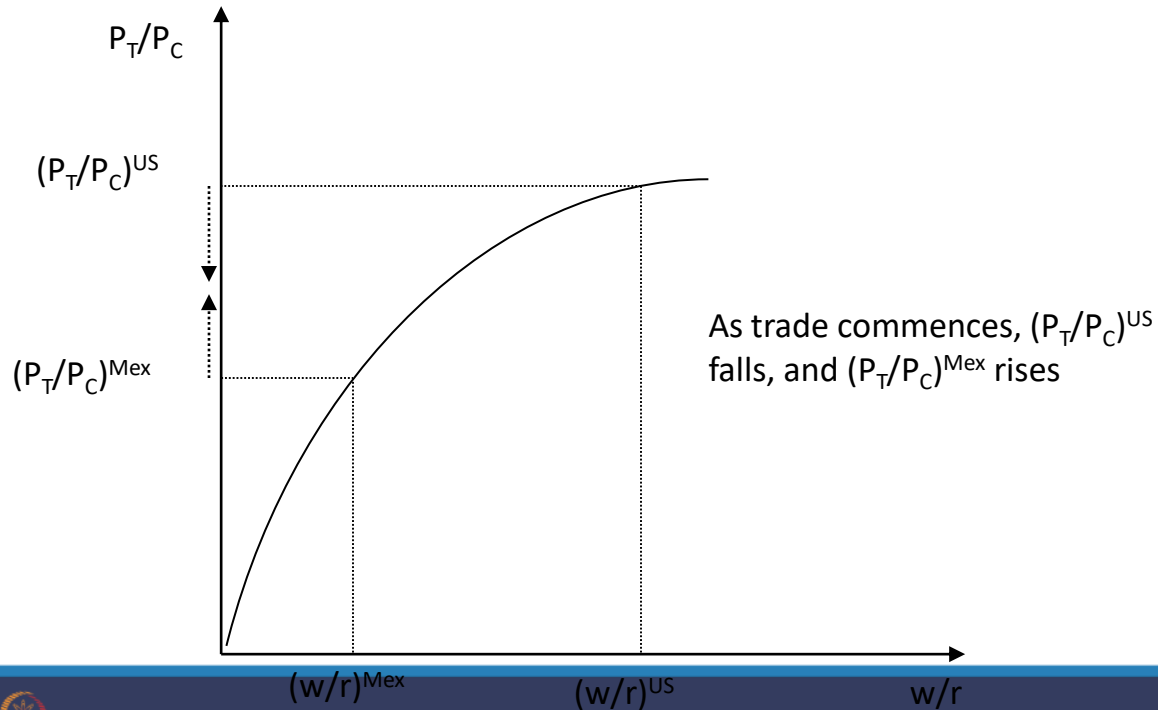
- The relatively capital abundant country (U.S.) exports the relatively capital intensive good (cars)
- The relatively labor abundant country (Mexico) exports the relatively labor intensive good (textiles)



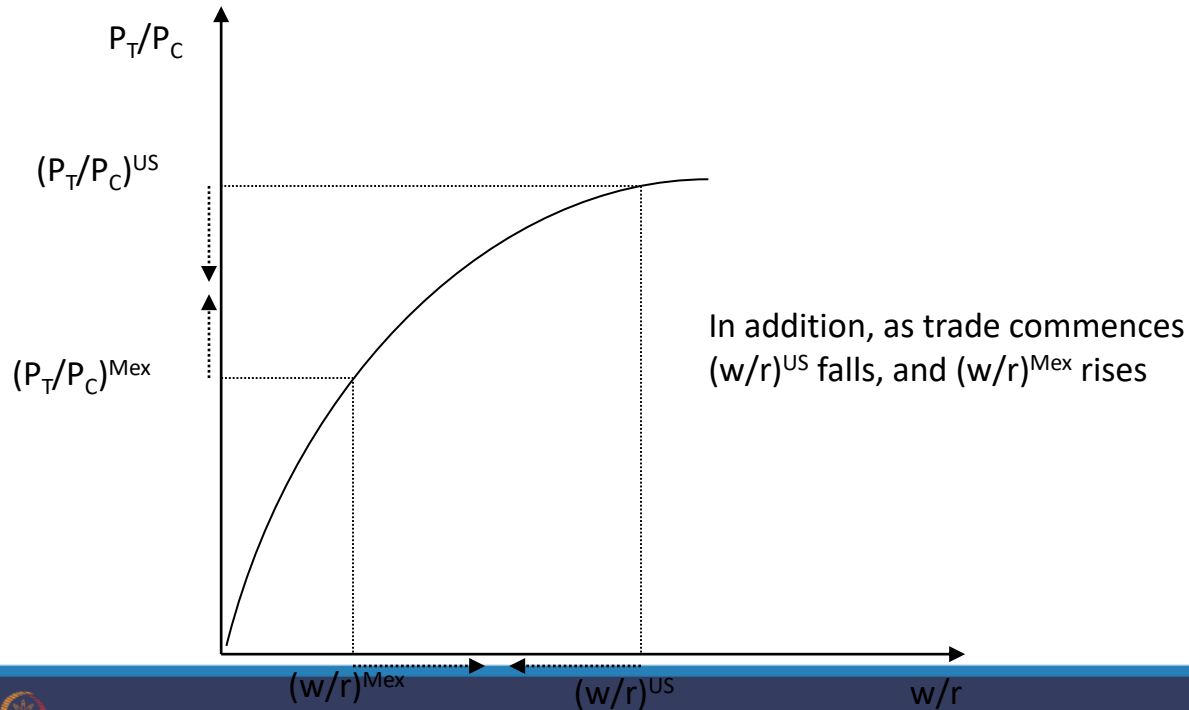
The Relative Cost Curve

- Since relative factor and commodity prices move together as countries move from autarky to trade, we can look at a picture of these two sorts of prices
- We also know that in autarky $(P_T/P_C)^{\text{Mex}}$ is lower than $(P_T/P_C)^{\text{US}}$
- $(w/r)^{\text{Mex}}$ is also lower in autarky than $(w/r)^{\text{US}}$ since Mexico is relatively labor abundant

The Relative Cost Curve



The Relative Cost Curve

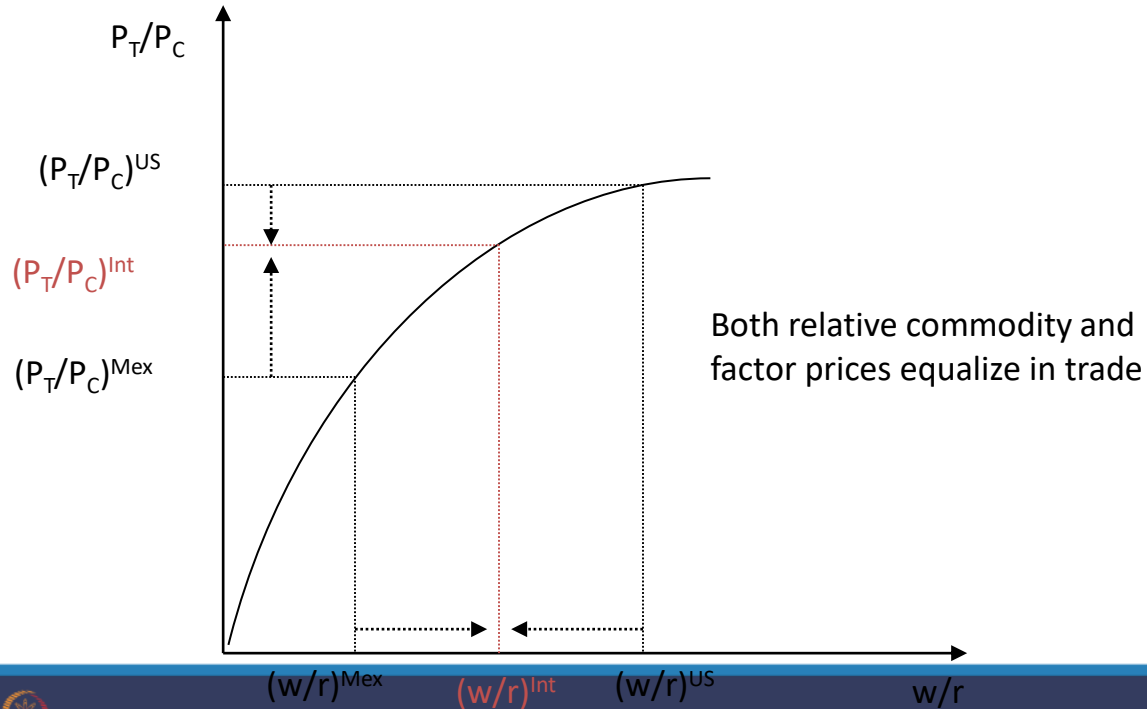


Factor Price Equalization

- In fact, each country moves along the relative cost curve to a common point
- We know this because there is only one international price
- If the countries wind up with a common international price, they must also wind up with a common set of relative factor prices



The Relative Cost Curve



The Factor Price Equalization Theorem

In equilibrium, with both countries facing the same relative (and absolute) product prices, both having the same technology, and with constant returns to scale, relative (and absolute) costs will be equalized. This can only happen if factor prices are equalized between countries.



H-O-S and the Distribution of Income

- The H-O theorem, together with the FPE theorem also tell us about how the incomes of different groups within a country change as trade starts
- This provides insight into the politics of free trade



The Stolper-Samuelson Theorem

*The increase in the price of the relatively **abundant factor** and the fall in the price of the relatively **scarce factor** because of trade implies that the owners of the abundant factor will find their real incomes rising; the owners of the scarce factor will find their real incomes falling.*



Stolper-Samuelson Theorem, Formal Statement, 1

- *Theorem:* Under the assumptions of the HOS model, an increase in the relative price of a good will raise the return to the factor used intensively in the production of that good *relative to all other prices*, and lower the return to the other factor, *relative to all other prices*.
- *American Case*

$$\hat{r} > \hat{P}_S > \hat{P}_B = 0 > \hat{w};$$

Stolper-Samuelson Theorem: Empirical Research

- Most of the empirical research suggests that trade, at least *via* Stolper-Samuelson channels, explains only a small amount of change in relative wages.
 - Most economists think *technological change* is a more important source of change in relative wages;
 - However, some economists think that other forms of globalization—e.g. *foreign direct investment, outsourcing, and effects on unions and welfare state*—may be very important.



All Factors Fixed: The Cairnes-Haberler Model

- In the very short-run, it seems reasonable to assume that virtually all factors of production are immobile.
- But this means that the proportions in production are fixed, so
 - Marginal physical products are fixed, so
 - If commodity prices are fixed factor payments will be fixed, and
 - If commodity price changes, the returns to all factors in an industry change by the same proportion as the price change.



Some factors fixed, others mobile: The Ricardo-Viner Model, 1

- The idea here is that some factors are mobile, while other factors are fixed.
 - *Land* is pretty fixed in its broad occupation;
 - *Capital* (i.e. *machines*, etc.) is also quite fixed;
 - *Labor* can be assumed mobile.
 - Are some kinds of labor more mobile than others?

Some factors fixed, others mobile: The Ricardo-Viner Model, 2

- What happens in the Ricardo-Viner model when relative commodity prices change?
- **Theorem:** An increase in the price of a good raises the return to the specific factor(s) used in the production of that good, and lowers the return to all other specific factors.
- **Neoclassical ambiguity:** the effect of relative commodity price changes on mobile factors is dependent on consumption shares.



General Adjustment Costs in an HOS World

- The Cairnes-Haberler and Ricardo-Viner models have the virtue of simplicity, but
- We can also consider a variant in which all factors are mobile, but at differential adjustment costs.
- The previous models are all variants of this model.
- For our purposes, there is little additional analytical or empirical gain from this generality.



Fixity and Mobility as Temporal Phenomena

- One way of thinking about the models we have been considering is to treat them as referring to different time horizons:
 - Cairnes-Haberler: very **short run**;
 - Ricardo-Viner: **medium run**; and
 - HOS: **long run**.
- Choosing models then will be related to the time horizon relevant for analysis.



Adjustment Costs, Implications

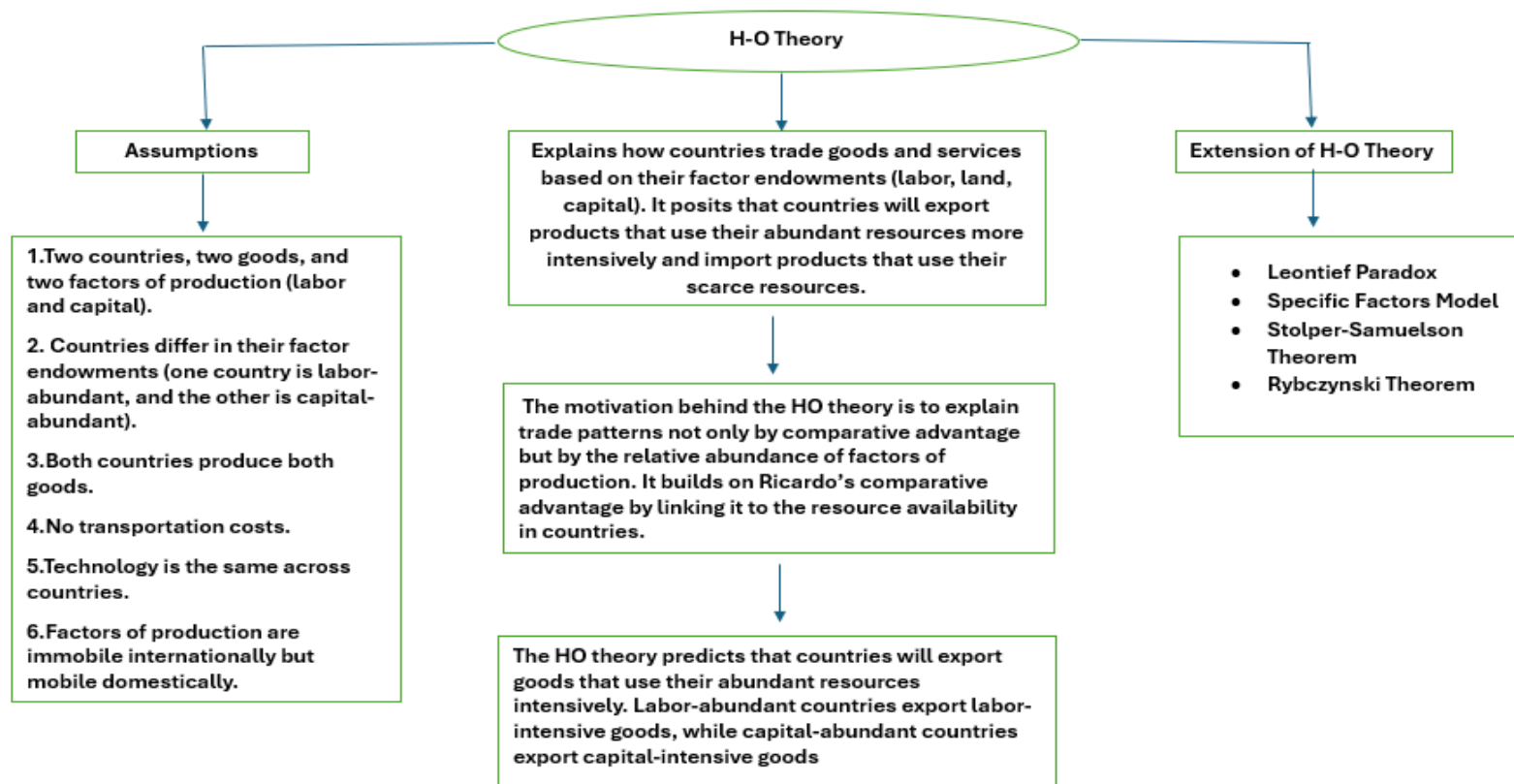
- *Static Gains from Trade are Overestimates*
- *Securing Gains from Trade Depends on Well-Constructed Trade-Adjustment Schemes*
 - Note that this is independent of long-run distributional effects.
 - Note, also, that this applies as much to increases in protection as to liberalization.
- *Securing Political Support for Liberalization May Also Depend on Trade-Adjustment Schemes*



Bottom Line on Demand Reversals

- If demand reversals occur, the H-O theorem no longer holds: the K-abundant country is exporting the L-intensive good, and the L-abundant country is exporting the K-intensive good
- If demand reversals are common in the real world, we'd better find a new theory





*Thank
you*

